QConfigMapping#

https://pytorch.org/docs/stable/generated/torch.ao.quantization.qconfig_mapp

Description:

Mapping from model ops to torch.ao.quantization.QConfig s. The user can specify QConfigs using the following methods (in increasing match priority): set_global : sets the global (default) QConfig set_object_type : sets the QConfig for a given module type, function, or method name set_module_name_regex : sets the QConfig for modules matching the given regex string

Function Signature

```
class
torch.ao.quantization.qconfig_mapping.QConfigMapping[source]
#

classmethod from_dict(qconfig_dict)[source]#

Return type

set_global(global_qconfig)[source]#

Return type
```

Returns:

dict[str, Any]

Code Examples

```
qconfig_mapping = QConfigMapping()
    .set_global(global_qconfig)
    .set_object_type(torch.nn.Linear, qconfig1)
    .set_object_type(torch.nn.ReLU, qconfig1)
    .set_module_name_regex("foo.*bar.*conv[0-9]+", qconfig1)
    .set_module_name_regex("foo.*", qconfig2)
    .set_module_name("module1", qconfig1)
    .set_module_name("module2", qconfig2)
    .set_module_name_object_type_order("foo.bar",
torch.nn.functional.linear, 0, qconfig3)
```

```
qconfig_mapping = QConfigMapping()
    .set_module_name_regex("foo.*bar.*conv[0-9]+", qconfig1)
    .set_module_name_regex("foo.*bar.*", qconfig2)
    .set_module_name_regex("foo.*", qconfig3)
```

Detailed Documentation

Documentation

Mapping from model ops to torch.ao.quantization.QConfig s.

The user can specify QConfigs using the following methods (in increasing match priority):

set_global : sets the global (default) QConfig

set_object_type : sets the QConfig for a given module type, function, or method name

set_module_name_regex : sets the QConfig for modules matching the given regex string

set_module_name : sets the QConfig for modules matching the given module name

set_module_name_object_type_order : sets the QConfig for modules matching a combination of the given module name, object type, and the index at which the module appears

Create a QConfigMapping from a dictionary with the following keys (all optional):

“” (for global QConfig)

“module_name_object_type_order”

The values of this dictionary are expected to be lists of tuples.

Set the global (default) QConfig.

Set the QConfig for modules matching the given module name. If the QConfig for an existing module name was already set, the new QConfig will override the old one.

Set the QConfig for modules matching a combination of the given module name, object type, and the index at which the module appears.

If the QConfig for an existing (module name, object type, index) was already set, the new QConfig will override the old one.

Set the QConfig for modules matching the given regex string.

Regexes will be matched in the order in which they are registered through this method. Thus, the caller should register more specific patterns first, e.g.:

In this example, “foo.bar.conv0” would match qconfig1, “foo.bar.linear” would match qconfig2, and “foo.baz.relu” would match qconfig3.

If the QConfig for an existing module name regex was already set, the new QConfig will override the old one while preserving the order in which the regexes were originally registered.

Set the QConfig for a given module type, function, or method name. If the QConfig for an existing object type was already set, the new QConfig will override the old one.

Convert this QConfigMapping to a dictionary with the following keys:

“” (for global QConfig)

“module_name_object_type_order”

The values of this dictionary are lists of tuples.